into more complex polymeric products. Radiation decomposition (*radiolysis*) of water caused evolution of hydrogen and oxygen gas and formation of hydrogen peroxide. Conversely, it was shown that water could be synthesized through irradiation of a mixture of H_2 and O_2. In 1911 S. Lind found that 1 g of radium exposed to air resulted in the production of 0.7 g of ozone per hour. By relating this *radiation yield* to the number of ions produced by the amount of radiation, Lind initiated the quantitative treatment of radiation-induced changes.

8.1. ENERGY TRANSFER

The chemical effects of radiation depend on the composition of matter and the amount of energy deposited by the radiation. In this section we consider only the energy transfer. For this purpose it is practical to divide high energy radiation into (1) charged particles (e^-, e^+, α, etc) and (2) uncharged particles (n) and electromagnetic radiation (γ). The latter produce recoil atomic ions, products of nuclear reactions and electrons as charged secondary ionizing particles. The terms direct and indirect ionizing radiation are often used for (1) and (2) respectively.

The amount of energy imparted to matter in a given volume is

$$E_{\text{imp}} = E_{\text{in}} + \Sigma Q - E_{\text{out}} \tag{8.1}$$

where E_{in} is the energy (excluding mass energy) of the radiation entering the volume, E_{out} is the energy of the radiation leaving the volume, ΣQ is the sum of all Q-values for nuclear transformation that have occurred in the volume. For a beam of charged particles $E_{\text{in}} = E_{\text{kin}}$; for γ-rays it is E_γ. If no nuclear transformations occur, $\Sigma Q = 0$. For neutrons which are captured and for radionuclides which decay in the absorber, $\Sigma Q > 0$; in the case of radionuclides already present in the absorber, $E_{\text{in}} = 0$.

8.1.1. Charged particles

We learned in the previous chapter that the energy of charged particles is absorbed mainly through ionization and atomic excitation. For positrons the annihilation process (at $E_{\text{kin}} \approx 0$) must be considered. For electrons of high kinetic energy bremsstrahlung must be taken into account. However, in the following text we simplify by neglecting annihilation and bremsstrahlung processes. The bremsstrahlung correction can be made with the aid of Figure 7.9 which gives the *average specific energy loss* of electrons through ionization and bremsstrahlung.

It has been found that the average energy, w, for the formation of an ion pair in gaseous material by charged particles is between 25 and 40 eV. For the same absorbing material it is fairly independent of the type of radiation and the energy. Table 8.1 lists values of w in some gases. The ionization potentials j of the gases are lower than the

Table 8.1 Ion pair formation energies for charged particles. All values in eV

Absorber	w	j	w-j
He(g)	43	24.5	18.5
H_2(g)	36	15.6	20.4
O_2(g)	31.5	12.5	19
Air	34	15	19
H_2O(g or l)	38	13	25
Ar(g) 5 MeV α	26.4		
Ar(g) 340 MeV p^+	25.5		
Ar(g) 10 keV e^-	26.4		
Ar(g) 1 MeV e^-	25.5		
Ar(g) average	26	15.7	10.3

typical *w*-values and thus the rest of the energy, $w-j$, appears as excitation energy. Since the excitation energies per atom are $\leq$5 eV, several excited atoms are formed for each ion pair formed. While it is easy to measure *w* in a gas, it is more difficult to obtain reliable values for liquids and solids. They also differ more widely; e.g. *w* is 1300 eV per ion pair in hexane (for high energy electrons) while it is about 5 eV per ion pair in inorganic solids.

The *specific energy loss* of a particle in matter is called the *stopping power* $\widehat{S}$,

$$\widehat{S} = \mathrm{d}E_{\mathrm{loss}}/\mathrm{d}x \ (\mathrm{J/m}) \tag{8.2}$$

where *x* is the distance traversed by the particle. To a good first approximation the stopping power of a material is determined by its atomic composition and is almost independent of the chemical binding of the atoms. Stopping power is a function of the particle velocity and changes as the particle is slowed down.

The *specific ionization J* is the number of ion pairs produced per unit path length

$$J = \mathrm{d}N_{\mathrm{j}}/\mathrm{d}x \,(\mathrm{ion\ pairs/m}) \tag{8.3}$$

The value of *J* depends on the particle and its energy as seen from Figure 7.7. The relation between $\widehat{S}$ and *J* is

$$\widehat{S} = w\,J \ (\mathrm{J/m}) \tag{8.4}$$

The *mass stopping power,* $\hat{S}/\rho$, is commonly expressed in units of MeV/g cm^2.

Another important concept is the *linear energy transfer* (abbreviated as *LET*) of charged particles. It is defined as the energy absorbed in matter per unit path length travelled by a charged particle

$$LET = \mathrm{d}E_{\mathrm{abs}}/\mathrm{d}x \tag{8.5}$$

Values of *LET* in water are given for various particles and energies in Table 7.2. For the same energy and the same absorbing material, the *LET* values increase in the order:

high energy electrons (also approximatively γ-rays
< β-particles (also approximately soft X-rays)
< protons
< deuterons
< heavy ions (ions of N, O, etc.)
< fission fragments

The relationship between *LET*, which refers to the energy absorbed in matter, and the stopping power, which refers to the energy loss of the particle, is

$$dE_{loss}/dx = dE_{abs}/dx + E_x \tag{8.6}$$

The difference E_x in these two energy terms is related to the energy loss by electromagnetic radiation (mainly bremsstrahlung).

8.1.2. Uncharged radiation

When neutrons or photons having the incident particle energy E_{in} are absorbed, a certain fraction of energy E_{tr} is transferred into kinetic energy of charged particles when traversing the distance dx. We can define an *energy transfer coefficient* as

$$\mu_{tr} = E_{in}^{-1} dE_{tr}/dx \left(m^{-1}\right) \tag{8.7}$$

If we neglect the bremsstrahlung associated with the absorption of the secondary charged particles formed in the initial absorption processes, E_{tr} is the energy absorbed (designated as E_{abs}), and we can write Eqn (8.7) as

$$\mu_a = E_{in}^{-1} dE_{abs}/dx \left(m^{-1}\right) \tag{8.8}$$

8.2. RADIATION TRACKS

The energy lost when a high energy charged particle is slowed in matter gives rise to a trail (or, more commonly, track) of ionized and excited molecules along the path of the particle; Figure 8.1 schematically depicts a track. Cloud chamber photographs of such tracks are shown in Figure 7.5. A photon imparts a large fraction of its energy to a single electron which subsequently ionizes and excites many other molecules along its path. The absorption of any type of ionizing radiation by matter thus leads to the formation of tracks of ionized and excited species. Whereas these species generally are the same in a particular sample of matter regardless of the type of radiation, the tracks may be sparsely or densely populated by these species. Expressions such as linear energy transfer (*LET*) and stopping power ($\hat{S}$) are based on the implicit assumption of a continuous slowing down process and thus gives quantitative information on the average energy loss but only qualitative information on the densities of reactive species.